

Emergency Surgery Or noT for common Vascular conditions in the periods before and during COVID-19 (ESORT-V): Study Protocol

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Summary of protocol amendments

Version 1.0: Original approved protocol

Version 2.0: Incorporates Amendment 1 (dated 12/06/2026)

Amendment 1 (12/06/2026)

Section reference	Original wording	Revised wording (changes bolded)	Rationale
Throughout	Acute limb ischaemia	Chronic limb-threatening ischaemia	There was a mistake in v1 of the protocol – TTE-2 is focused on cases of chronic limb-ischaemia rather than acute.
Target trial emulation – Assignment procedures and follow-up (main text) and corresponding sections in Table 1	“We followed the advice of the clinical panel in defining time zero as the threshold date (the date at which the AAA met the threshold for intervention) in TTE-1, the referral date (the date at which the patient was referred for lower limb bypass) in TTE-2A, or the admission date (the date at which a patient has a non-elective admission requiring lower limb bypass surgery) in TTE-2B. These dates are all recorded in the NVR database (Table 1). Follow-up will then begin on these same dates (Figure 1).”	“We followed the advice of the clinical panel in defining time zero as the threshold date (the date at which the AAA met the threshold for intervention) in TTE-1, the “qualifying” out-patient visit dates in HES (as per Li et al 2022¹) in TTE-2A, or the admission date (the date at which a patient has a non-elective admission requiring lower limb bypass surgery) in TTE-2B. We could not use the NVR referral date as time zero in TTE-2A as originally planned due to the high levels of missingness for this particular variable. For TTE-2A and TTE-2B, eligibility criteria and baseline covariates were defined at time zero. Treatment strategy was defined by the interval between time zero and lower-limb bypass surgery, however, clinical outcomes were calculated from the date of surgery. ”	The referral date for chronic limb ischaemia interventions captured in the NVR had substantial levels of missingness consistent across vascular centres in England, which was not anticipated at the study design stage. We follow precedent research in the NVR in using the out-patient visit date instead. For TTE-2A and TTE-2B we measured outcomes from the date of surgery to ensure that the outcome reflected post-operative recovery and was not driven by differences in the time between time zero and surgery.
Outcomes (main text) and	“Secondary outcomes will be DAOH up to 1-year follow-up, mortality at 1-year	“Secondary outcomes will be DAOH up to 1-year follow-up, total length of stay and mortality up to 1-year follow-	Amputation-free survival is a common outcome in randomised controlled trials for chronic limb

corresponding sections in Table 1	follow-up, re-admissions, and re-interventions (including emergency and elective surgery)."	up, and re-admissions. In addition, for TTE-2, we will include re-vascularisation (index limb) and major amputation (index limb) as a binary event up to 90-, 180-, and 365-days. We will also include amputation-free survival, defined as the time from surgery to the first occurrence of index-limb major amputation, death, or administrative censoring at the study end date."	ischaemia, ^{2,3} and has been used as a primary endpoint in precedent observational research. ⁴ This outcome is of clinical interest and has been added prior to any adjusted analyses.
Subgroups and corresponding sections in Table 1	"We will report the clinical effectiveness overall, and according to pre-specified subgroups. These include age, sex, deprivation, refined diagnostic group (e.g., aneurysm diameter, severity of limb ischaemia, prior attempts at revascularisation), comorbidities, frailty measured using the SCARF index ⁵ derived from the HES admission mapped to the NVR index admission, and past medical history (Table 1)."	"We will report the clinical effectiveness overall, and according to pre-specified subgroups. These include age, sex, deprivation, refined diagnostic group (e.g., aneurysm diameter, severity of limb ischaemia, prior attempts at revascularisation), comorbidities (in particular diabetes mellitus), frailty measured using the SCARF index ⁵ derived from the HES admission mapped to the NVR index admission, and past medical history (Table 1). For TTE-2 (chronic limb-threatening ischaemia), we will also report on tissue loss versus rest pain as defined by the Fontaine score recorded in the NVR (nocturnal and/or resting pain vs necrosis and/or gangrene in the limb)."	The additional subgroups added for TTE-2 are of particular clinical and policy interest. We have added these subgroups to the pre-specified subgroup list before conducting any adjusted analyses.
Clinical effectiveness analyses	"The first-stage model estimates the propensity for type of surgery (early versus later) conditional on the instrument and measured confounders using a logit or probit model... These models will include measured confounders (including measured contextual confounders which may confound the	"The first-stage model estimates the propensity for type of surgery (early versus later) conditional on the instrument and measured confounders using a logit or probit model... These models will include measured confounders (including measured contextual confounders which may confound the instrument and outcome) as has been done in previous	The protocol specified adjustment for measured confounders but did not specify how the final covariate set would be selected. A LASSO-based procedure was added as an optional model-specification step to support parsimonious adjustment when the number of candidate covariates and interactions was large

	instrument and outcome) as has been done in previous instrumental variable analyses.”	instrumental variable analyses. Where appropriate, the final covariate set, including main effects and interactions, will be determined by a LASSO-based selection procedure applied to both the first-stage and second-stage outcome models.”	relative to the available data.
Alternative analyses (main text) and corresponding sections in Table 1	“We will also consider alternative cutoffs for the definitions of ‘early’ or ‘later’ surgery, for example for TTE-1 we will limit the window for ‘later’ surgery to 57–84 days (vs 57–150, base case).”	“We will also consider alternative cutoffs for the definitions of ‘early’ or ‘later’ surgery. For TTE-1, we will redefine early surgery as 15–84 days and later surgery as 85–150 days after the threshold date. For TTE-2A, we will consider a narrower definition of later surgery as lower limb bypass on days 15–28 from the qualifying outpatient visit, compared with days 15–60 in the base case.”	These sensitivity analyses were added following clinical panel input to assess whether findings were robust to alternative timing windows. For TTE-1, the definition of early surgery was extended from 15–56 days after the threshold date in the main analysis to 15–84 days, based on the NAAASP definition of a “long wait” for AAA repair. For TTE-2A, later surgery was redefined as 15–28 days after the qualifying outpatient visit, rather than 15–60 days, to assess sensitivity to a narrower elective treatment window and to reduce the possibility that longer waiting periods reflected interval clinical deterioration or a shift away from the original elective context.

DAOH: days alive and out of hospital; HES: Hospital Episode Statistics; NVR: National Vascular Registry; TTE: target trial emulation;

Abstract

Introduction

Uncertainty exists around the optimal timing of most vascular surgery operations. Clinical guidelines suggest delivering urgent surgery, but there is little evidence demonstrating this approach is more clinically and cost effective than later surgery, which can allow time for patients to prepare for major surgery. The delivery of rapid surgery poses significant service delivery challenges, and many vascular centres in the UK National Health Service (NHS) do not meet targets for surgical delivery. This challenge of meeting criteria was compounded by the COVID-19 pandemic.

The aim of the Emergency Surgery Or noT for common Vascular conditions study (ESORT-V) is to provide a rigorous evaluation of the relative clinical and cost-effectiveness of 'early' or 'later' surgery for patients with common vascular conditions, including abdominal aortic aneurysms (AAA) and peripheral arterial disease (PAD). This study recognises the impact of the COVID-19 pandemic on vascular surgery provision. ESORT-V was commissioned by the National Institute for Health Research for Patient Benefit (NIHR RfPB) programme.

Methods and Analysis

The National Vascular Registry (NVR), Hospital Episode Statistics (HES), and Office for National Statistics (ONS) linked data will be used to identify patients with non-ruptured AAA undergoing AAA repair and PAD undergoing lower limb revascularisation between January 2014 to December 2023. We will combine an emulated target trial with an instrumental variable analysis to minimise the risk of bias, including unmeasured confounding. Our instrumental variable will be each vascular centre's 'tendency to expedite surgery', defined as the propensity to perform surgery before (early) or after (later) defined timepoints. The primary outcome will be days alive and out of hospital. Incremental cost-effectiveness will be reported by assessing the incremental net monetary benefits of early versus later surgery at 1 year follow-up. We will report clinical and cost-effectiveness overall and for pre-defined subgroups.

Ethics & Dissemination

The study is approved by the LSHTM research ethics committee (project ID 27065). Data sharing agreements have been obtained with National Health Service Digital and the NVR. Results will be disseminated via NIHR RfPB reporting, peer-reviewed journals, and conferences.

Strengths and Limitations of this Study

- The ESORT-V study will consider all eligible early and later vascular surgical operations across the UK over a 5-year period in a comparative clinical and cost-effectiveness analysis.
- This study will use the target trial emulation framework combined with an instrumental variables analysis to reduce the risk of biases common in observational studies, including residual confounding.
- This study will involve clinical and patient and public involvement (PPI) panels to guide translation of results into clinical practice, and inform the design of future randomised-controlled trials.
- The instrumental variable analyses are subject to assumptions which cannot be fully tested.

Introduction

Each year, approximately 40,000 vascular surgery operations are carried out in England to treat arterial diseases, including abdominal aortic aneurysms (AAA) and peripheral arterial disease (PAD).⁶ Vascular surgery is often performed on people living with multiple long-term conditions and older age groups, contributing to high complication rates, re-admissions, and mortality.^{7,8} Consequently vascular surgery is associated with significant costs and resource utilisation for the NHS.⁹

The NHS Getting It Right The First Time (GIRFT) report in 2018 recognised the need to reconfigure all vascular surgery as ‘urgent’ to reflect clinical need and reduce between-centre variation in waiting times.¹⁰ However, there is a lack of evidence to inform decisions on the timing of surgery for common vascular procedures such as AAA repair and lower limb bypass. Although various waiting time targets exist, many centres in the UK do not meet them. For AAA repairs, an 8-week target has been recommended by the National AAA Screening Programme (NAAASP),¹¹ but the median delay from assessment to AAA repair was 50 to 160 days. In severe limb ischaemia, the evidence that urgent limb salvage improves outcomes is weak.¹²

The challenges of service provision for urgent vascular surgery was further compounded by the COVID-19 pandemic, where operating capacity was severely reduced, leading to cancellations and delays.¹³ In 2021, approximately 11,000 elective and 12,000 emergency vascular operations were performed for AAA repair and lower limb revascularisation.¹⁴ These numbers were lower than previous years, particularly for AAA repair. It is anticipated that in the medium to long term, surgical capacity in the NHS will be reduced.¹⁵ NHS England and GIRFT guidance has encouraged deferral of surgery, adjusted thresholds for intervention, and increased use of temporising non-operative management.¹⁶

The current guideline thresholds for urgent vascular surgery may also limit proper prehabilitation and risk modification prior to surgery. Substantial improvements have been made in peri-operative care and medical optimisation in recent years, which may significantly improve surgical outcomes despite delaying the timing of surgery.¹⁷ There is some evidence that very few AAAs rupture whilst awaiting surgery,¹¹ and other randomised trials have identified patient benefit from optimising comorbidities and fitness before AAA surgery.^{18,19} Nevertheless, emergency repair of ruptured AAAs have a very high mortality of 44.2%, with only two-thirds being discharged home.^{9,14} There is therefore a pressing need for evidence comparing ‘urgent’, which we define as ‘early’, versus ‘elective’, which we shall define as ‘later’, vascular surgery for conditions like AAA. This research will directly address the GIRFT recommendations, as well as the James Lind Alliance perioperative care research priorities,²⁰ to inform the surgical management of patients with common vascular conditions. This research has direct implications for patients and the organisation and delivery of health services.

The Emergency Surgery or Not (ESORT) study previously found that for five acute general surgical conditions, emergency surgery and non-emergency strategies led to similar average days alive

and out of hospital (DAOH) at 90 days, but may be modified by individual patient characteristics. This highlights the need for further research to optimise emergency surgery provision.²¹

The NVR is an audit of vascular surgery procedural services delivered at NHS hospitals in England. These routinely collected health data offer a unique opportunity to compare clinical outcomes and costs to the NHS between patients who receive ‘early’ versus ‘later’ vascular surgery. However, a major limitation of these observational data is that they will be subject to bias common in observational studies – particularly confounding bias by unmeasured confounders. The Target Trial Emulation (TTE) framework is a helpful tool that can be used to reduce the risk of common biases in observational studies seeking to infer causal effects from observational data.^{22,23} We will use the TTE framework to design this study, which will also incorporate an instrumental variable analysis to reduce the risk of bias from unmeasured confounding.^{22,24}

The aim of the Emergency Surgery Or noT for common Vascular conditions study (ESORT-V) study is to provide a rigorous evaluation of the relative clinical and cost-effectiveness of ‘early’ versus ‘later’ surgery for patients with common vascular conditions, including AAA and PAD, recognising the impact of the COVID-19 pandemic on vascular surgery provision. This research study is funded by the National Institute for Health Research for Patient Benefit programme (NIHR RfPB). The specific objectives of the study are:

- To assess the clinical and cost-effectiveness of ‘early’ versus ‘later’ surgery for AAA repair (TTE-1) and lower limb bypass (TTE-2).
- To assess the clinical and cost-effectiveness of ‘early’ versus ‘later’ surgery for both TTEs across different patient subgroups and time periods defined before, during and after the COVID-19 pandemic.
- To provide recommendations about which subgroups should have ‘early’ versus ‘later’ surgery and define sub-populations and/or comparison groups as priorities for future RCTs.

Methods and Analysis

Overview

The ESORT-V study will use previously collected routine data in cohort studies which emulate “target trials” to estimate the clinical and cost-effectiveness of ‘early’ versus ‘later’ surgery for common vascular conditions in the context of the COVID-19 pandemic. A clinical panel will help to refine the target populations and interventions, as well as finalising the definition of ‘early’ and ‘later’ surgery. A patient and public involvement (PPI) panel, including patients with lived experience of these vascular conditions, will advise on the magnitude of difference in the main clinical endpoints that is important to patients. We will use the TTE framework in the design and reporting of ESORT-V.^{22,23}

Data Source

NVR data²⁵ will be linked with the Hospital Episodes Statistics (HES)²⁶ and the Office for National Statistics (ONS) databases to identify the population of interest and relevant outcomes. We will define patient cohorts admitted between January 2015 to December 2023 for non-ruptured AAA undergoing AAA repair (TTE-1) or PAD undergoing lower limb revascularisation (TTE-2).

The NVR is managed by the Clinical Effectiveness Unit of the Royal College of Surgeons of England as a national clinical audit. It has previously been shown to capture 95% of vascular surgery cases and has a median of 93.3% data completeness with external and internal validation.²⁷ The lower limb angioplasty or stenting cohort is the only group limited by lower case ascertainment rates. For this cohort, the main analysis will limit eligibility to those vascular centres that exceed the 85% case ascertainment threshold as set by GIRFT, which will include approximately 60% of eligible cases. Sensitivity analysis will consider using alternative thresholds for inclusion (80% or 90%) to examine the extent to which these cases and centres are representative of all available cases and centres.

Patient demographics, admission details such as procedural type and admission method, information related to the pre-operative pathway, anaesthetic details, indications, risk scoring, and variables relating to AAA or bypass operations will be extracted from the NVR database, as well as post-operative and follow-up information. The included procedures are defined according to the Office of Population Censuses and Surveys (OPCS) codes, which is well-established in the NVR. Data from the HES and ONS database will inform patient demographics, comorbidities, mortality, and resource use data.

Target trial emulation

We designed the ESORT-V TTEs in accordance with the TARGET statement, which offers guidance for designing and reporting observational studies estimating causal effects by emulating a target trial.²⁸ We summarise the TTEs in **Table 1**, illustrate the study design using a causal diagram in **Figure 1**, and provide detail in the following sections.

Eligibility criteria

It is necessary to clearly specify the eligibility criteria for the population of interest and subgroups for whom there is clinical uncertainty as to the potential benefits for ‘early’ versus ‘later’ surgery. The target population and treatment strategies, including the definitions of ‘early’ and ‘later’ surgery were informed by consensus of a clinical panel as part of an online design workshop. The panel consisted of 18 panellists with expertise in vascular surgical conditions based in vascular centres across the UK, including surgeons, physicians, interventional radiologists, anaesthetists, and specialist nurses.

In TTE-1, we will focus on people with asymptomatic AAA who underwent either open or endovascular repair (EVAR) surgeries between 2015 to 2023. These people must be eligible for linkage to HES data. Exclusion criteria include having symptomatic AAA, having previously

undergone surgery for AAA, undergoing complex EVAR, revision open, revision EVAR, endovascular aneurysm sealing (EVAR), had a AAA diameter >70mm, underwent surgery in ≤ 14 days or >150 days from the threshold date (the date AAA reached threshold for intervention), or had active/managed cancer at the time of surgery. These exclusions were made in consultation with clinical colleagues to capture the patient population for which there is genuine clinical uncertainty in the relative effectiveness of ‘early’ versus ‘late’ surgery to repair AAA.

In TTE-2, we will focus on people (A) undergoing lower limb bypass admitted to hospital electively, and (B) undergoing lower limb bypass admitted to hospital non-electively. We will include those with peripheral artery disease (PAD) including tissue loss, infection, and gangrene between 2015 to 2023. Those included must be eligible for HES linkage. Exclusion criteria include having a Fontaine score <3, an American Society of Anesthesiologists (ASA) grade 5, an aneurysm, trauma, uncontrolled infection, or neuropathy. We will also exclude patients whose surgery is at a low volume centre.

Treatment strategies

We developed an initial definition of ‘early’ and ‘later’ surgery from national clinical guidance, which we then refined through discussion with our clinical panel.^{10,11,29-32}

In TTE-1, we will define ‘early’ surgery as expedited surgery to repair AAA which is 15 to 56 days after the threshold date. ‘Later’ surgery will be defined as elective (delayed) surgery to repair AAA from 57 to 150 days after the threshold date.

In TTE-2A (elective lower limb bypass), we will define ‘early’ surgery as lower limb bypass on days 1-14 from a “qualifying” out-patient visit date (defined as per Li et al 2022¹), and ‘later’ surgery will be defined as lower limb bypass on days 15-60 from the qualifying out-patient visit date. In TTE-2B (non-elective lower limb bypass), we will define ‘early’ surgery as lower limb bypass on days 1-5 from the admission date, and ‘later’ surgery as lower limb bypass on days 6-21 from the admission date. The clinical panels consensus choice of cutoffs recognised that those patients who had surgery prior to the start of these time windows or after the last day were likely to differ in their prognosis. The consensus of the panel was that protocols for the ideal RCT would not include ‘earlier’ or ‘later’ times for either of the TTEs (see also alternative analyses).

Assignment procedures and follow-up

‘Time zero’ refers to the ‘point of randomisation/treatment assignment’ in a target trial emulation. Carefully defining time zero is important to minimise the risk of bias, particularly immortal time bias.^{22,33} In an RCT, time zero is defined as the time when patients have met the eligibility criteria, treatment strategies are assigned, and follow-up begins. In observational studies, these 3 time points can be difficult to align due to limited or missing data capture.

In ESORT-V, the choice of time zero is not straightforward as this can depend on the mode of admission, particularly for lower limb surgery.¹ The choice of time zero is also challenging for conditions where patients may be under surveillance for a long time before a decision is made to operate (e.g., in AAA repair), or where surgery is delayed for unforeseen reasons (e.g., if the patient became medically unfit). Clinical panel discussion also identified challenges related to

differences in patient pathways and how this could impact identification of time zero (for example, planned elective admission versus urgent admission eventually leading to surgery).

We followed the advice of the clinical panel in defining time zero as the threshold date (the date at which the AAA met the threshold for intervention) recorded in the NVR in TTE-1, the “qualifying” out-patient visit date recorded in HES (as per Li et al 2022) in TTE-2A, and the admission date (the date at which a patient has a non-elective admission requiring lower limb bypass surgery) recorded in the NVR in TTE-2B. We could not use the NVR referral date as time zero in TTE-2A as originally planned due to the high levels of missingness for this particular variable. For TTE-2A and TTE-2B, eligibility criteria and baseline covariates were defined at time zero. Treatment strategy was defined by the interval between time zero and lower-limb bypass surgery, however, clinical outcomes were calculated from the date of surgery. We acknowledge the potential for immortal time bias,³⁴ since the NVR excludes people who are eligible but do not have vascular surgery. We also could not align eligibility, treatment assignment, and start of follow-up. However, deaths are uncommon between the threshold date and surgery in TTE-1 (asymptomatic AAA repair) and TTE-2A (elective lower limb revascularisation). There is greatest potential for immortal time bias in TTE-2B (non-elective lower limb revascularisation), which we will incorporate into our reporting and interpretation. We cannot use clone-censor-weighting^{33,35} to account for this immortal time bias here because the NVR does not include those people who die before receiving vascular surgery.

Outcomes

The primary outcome will be days alive and out of hospital (DAOH) up to 180 days after AAA repair (TTE-1) and up to 90 days after lower limb revascularisation (TTE-2). DAOH is a composite of mortality and total length of hospital stay, including re-admissions for re-interventions during the follow-up time period. This has been recognised as a patient-centred outcome measure and has been previously recommended and used in perioperative clinical trials.³⁶ It is considered a superior measure of surgery quality and perioperative care to reporting complication and mortality rates.³⁷ This outcome measure has been discussed and supported as an important outcome from previous PPI workshops as part of the previous ESORT study.²¹ There are two common forms of DAOH that differ in the calculation of DAOH: the first values the number of days out of hospital for those who die before the end of follow-up, the second assigns zero DAOH for these individuals who die during follow-up. We will use the first measure of DAOH in the main analysis (base case), and the second in a sensitivity analysis.

Secondary outcomes will be DAOH up to 1-year follow-up, total length of stay and mortality up to 1-year follow-up, and re-admissions. In addition, for TTE-2, we will include re-vascularisation (index limb) and major amputation (index limb) as a binary event up to 90-, 180-, and 365-days. We will also include amputation-free survival, defined as the time from surgery to the first occurrence of index-limb major amputation, death, or administrative censoring at the study end date.

Causal Contrasts

Our ideal trial would record at time zero (the threshold (TTE-1) or referral (TTE-2) date) *the intention* to have either early or later surgical strategies, and then follow-up all individuals irrespective of whether or not they received their intended strategies. There are 2 major challenges in the emulation of this ideal trial from the linked NVR-HES-ONS data. The first being that the intention to receive either strategy is not recorded, hence we cannot recognise that some patients may ‘crossover’ from their intended strategy. Second, the NVR only includes information on those who have surgery, and so the TTEs exclude patients who receive neither form of surgery, which may occur due to patient choice, change in clinical condition, or death.³⁸

Hence, we cannot estimate the observational analogue of the intention-to-treat (ITT) effect. We will therefore report the per protocol (as received) effect of ‘early’ versus ‘later’ surgery. For the per protocol estimand, we will report the average treatment effect across the overall study population, as well as conditional average treatment effects according to pre-specified subgroups. We will report the mean difference in DAOH and the risk differences for death on the absolute scale between treatment groups.

Identifying assumptions

The target trial assumes exchangeability (the treatment groups are comparable on average) and positivity (each patient has a non-zero probability of being assigned to the treatment or comparator). Randomisation to treatment ensures both of these assumptions are met.

Our TTEs makes these same assumptions of exchangeability and positivity. However, we cannot randomise since we are using observational data. We will use an instrumental variable analysis to avoid making the common assumption of exchangeability conditional on measured confounders, which is unlikely to hold in the vascular surgery setting. Instead we will exploit exogenous variation in the timing of vascular surgery across vascular centres in England as an instrument to conduct an instrumental variable analysis. The rationale for this instrumental variable analysis builds upon our previous work that used the ‘tendency to operate’ as an instrument to balance observed and unobserved prognostic measures between two comparison groups.^{22,24} This ‘tendency to operate’ was shown to vary widely across hospitals in England, but is not correlated with surgical skill or experience.³⁹ More specifically, in ESORT-V, we will use the ‘tendency to expedite surgery’ as the instrument in these TTEs, which we anticipate will vary substantially across vascular centres in England due to factors such as centre-level policies and surgeons prior beliefs, which are independent of the individual’s prognostic characteristics or the quality of surgical care in the centres. We define the ‘tendency to expedite surgery’ as the propensity of a unit to perform surgery before (early) or after (later) the defined timepoints. This will be calculated for each individual in the study population by taking the proportion of patients within the same vascular centre receiving early surgery in the 12-months prior to the individual’s time zero date.

We recognise that the instrumental variable makes 4 key assumptions which can only be partially tested (**Table 2**). We will investigate the validity of the instrumental variable in our statistical

analyses. We will also consider whether changing the lookback window (e.g., 6-months instead of 12-months) improves the validity of the instrument.

Statistical analysis

Descriptive analysis

Our descriptive analyses will focus on comparing measured confounders (**Table 3**) between treatment groups in both TTEs and ensuring there is suitable variation in the instrument to conduct an instrumental variable analysis in the clinical and cost-effectiveness analyses. We will describe health resource use before and during the treatment windows (e.g., length of hospital stay), since it is possible that people are hospitalised for variable periods prior to receiving either ‘early’ or ‘later’ vascular surgery.

We will describe variation in the instrument across hospitals overall and by study year to understand how the ‘tendency to operate’ varies across vascular centres in England for our conditions of interest.

Checking the IV assumptions

Before proceeding with the clinical and cost-effectiveness analyses, we will evaluate the instrumental variable assumptions (**Table 2**). To test the relevance assumption, we will report the strength of association between the instrument and the procedure received, conditional on measured confounders (**Table 3**). We will summarise the strength of association using the Kleibergen-Paap F-statistic⁴⁰ calculated from the first-stage model (see Clinical effectiveness analysis), which should have a value >100 to be judged a sufficiently strong instrument.⁴¹

We will also describe any association (or lack thereof) between deciles of the instrument and standardised values of the measured confounders across the study population. A lack of association between the instrument and measured confounders will provide some support for the assumption that the instrument is not associated with unmeasured confounders (assumption 2). We will also consider external evidence about whether there is support for the exclusion restriction (assumption 3) or monotonicity assumptions (4).

Clinical effectiveness analysis

In the main (base case) analysis, we will use the 2-stage residual inclusion (2SRI) instrumental variable model that can report an average treatment effect across the whole population of interest.⁴² The first-stage model estimates the propensity for type of surgery (early versus later) conditional on the instrument and measured confounders using a logit or probit model. The generalised residuals from this model are then included as additional covariates in the second-stage outcome model. For count data (e.g., DAOH), the outcome analysis will use generalised linear models (e.g., Tobit model⁴³). Probit or logit models will be used for binary outcomes (e.g., emergency admission) and survival analyses will use survival models (e.g., Cox proportional hazards model). These models will include measured confounders (including measured

contextual confounders which may confound the instrument and outcome) as has been done in previous instrumental variable analyses.⁴⁴ Where appropriate, the final covariate set, including main effects and interactions, will be determined by a LASSO-based selection procedure applied to both the first-stage and second-stage outcome models. This approach will improve precision, as well as account for any potential associations between the instrument and other contextual confounders like the year of surgery or COVID-19 pandemic time period, which could potentially be associated with the tendency to expedite surgery (i.e., the instrument) (**Figure 1**).

The absolute mean difference will be reported for DAOH, and absolute risk difference for binary endpoint measures. We will report the clinical effectiveness overall, and according to pre-specified subgroups. These include age, sex, deprivation, refined diagnostic group (e.g., aneurysm diameter, severity of limb ischaemia, prior attempts at revascularisation), comorbidities (in particular diabetes mellitus), frailty measured using the SCARF index⁵ derived from the HES admission mapped to the NVR index admission, and past medical history (**Table 1**). For TTE-2 (chronic limb-threatening ischaemia), we will also report on tissue loss versus rest pain as defined by the Fontaine score recorded in the NVR (nocturnal and/or resting pain vs necrosis and/or gangrene in the limb). The clinical and cost-effectiveness of urgent versus elective surgery will also be reported for periods prior to and after the onset of the COVID-19 pandemic. Specifically, we will report treatment effects using a broad definition of the COVID-19 pandemic: pre-COVID-19 (study start to 2020 March 15); COVID-19 (2020 March 16 to 2022 March 11); and post-COVID-19 (2022 March 12 to study end). We will consider a finer categorisation of COVID-19 time period; however, we consider it unlikely that the study population will be large enough to draw meaningful conclusions from this finer categorisation: pre-COVID-19 (study start to 2020 March 15); Wave 1 (2020 March 16 to 2020 May 12); Post-Wave 1 (2020 May 13 to 2020 November 3); Wave 2 (2020 November 4 to 2021 March 11); Post-Wave 2 (2021 March 12 to 2021 November 3); Wave 3 (2021 November 4 to 2022 March 11); Post-Wave 3 (2022 March 12 to study end).⁴⁵

We will consider alternative analyses to evaluate the clinical effectiveness overall and according to pre-specified subgroups. This will include unsupervised approaches to assess treatment effect heterogeneity across patient phenotypes.⁴⁶

Sample Size Calculation

Following the recommendations for the design of IV studies,²⁴ power calculations were performed under different assumptions, allowing the minimal clinically important difference (MCID) to range from 3 to 6 days DAOH before 180 (TTE-1) or 90 (TTE-2) days, and allowing for differences in strengths of IV (70% or 80% compliance). The number of eligible procedures will be at least 12,000 AAA repairs and 20,000 lower limb revascularisations. For both conditions, the study will have at least 80% power to detect differences between the comparison groups in the mean DAOH that are as small as 3 days, at a 5% level of statistical significance.

Cost-Effectiveness Analysis

The cost-effectiveness analysis will take a hospital and personal health services perspective⁴⁷ over a time horizon of 1-year (base case). We will report patient-level resource use extracted from

the NVR-HES linked data for all admissions and readmissions up to 1 year. Unit costs will be obtained from the NHS payment by results database⁴⁸ and the personal health services perspective research unit (PSSRU) unit cost database.⁴⁹ These unit costs will then be combined with resource use measures to report total costs per patient up to 1 year (TTE-1 and -2).

In addition to total life years, we will report quality adjusted life years (QALYs) up to 1-year. We will identify health-related quality of life (HrQoL) estimates from the literature and guidelines,⁵⁰ using EQ-5D as the generic instrument. QALYs will then be calculated by combining each person's survival time with appropriate HrQoL using the area-under-the curve approach.⁵¹

We will report incremental net monetary benefits (INBs) of the early versus later surgery groups by multiplying the incremental QALYs according to NICE-recommended willingness-to-pay threshold (£20,000 to £30,000 per QALY) and subtracting the total costs. We will use the same instrumental variable analysis as in the clinical effectiveness analysis to estimate the treatment effect for the cost-effectiveness analysis outcomes (incremental costs, incremental QALYs, and incremental net monetary benefit).

In the sensitivity analysis, we will evaluate health economic outcomes using a 3-year follow-up period.

Alternative analyses

We will assess the extent to which the overall clinical and cost-effectiveness results differ according to alternative standpoints to those taken in the main analysis. For example, we will consider the alternative version of the DAOH that does not value days out of hospital for those who die before the end of follow-up. We will also consider alternative cutoffs for the definitions of 'early' or 'later' surgery. For TTE-1, we will redefine early surgery as 15–84 days and later surgery as 85–150 days after the threshold date. For TTE-2A, we will consider a narrower definition of later surgery as lower limb bypass on days 15–28 from the qualifying outpatient visit, compared with days 15–60 in the base case. We will apply analytical methods (e.g. weighted regression) that do not rely on the validity of the instrumental variable but instead assume there are no unobserved confounders. In the economic evaluation we will consider costs and outcomes over an extended time horizon (e.g., 3 years).

Patient & Public Involvement

We will host a PPI workshop to encourage active contribution to the interpretation and dissemination of results by clinicians and patients with lived experience. In this workshop, we will discuss the best strategy to present evidence generated on the relative benefits, risks, and costs to patients of early versus delayed surgical strategies. The study design will recognise that an important research output will be to inform the design of a future RCT.

The PPI translation workshop will be held to discuss the findings from the ESORT-V study, and to discuss the merits and drawbacks of alternative trial designs. This will ensure that the proposed RCT can answer questions which are important to patients with lived experience of the vascular

conditions concerned, vascular surgeons, and service commissioners. The clinical panel will advise on those conditions and sub-conditions in ESORT-V, for which there is sufficient equipoise and willingness to enrol patients in any subsequent RCT. The PPI group will confirm that patients are willing to be randomised and participate in such research, and refine the outcome measures of interest.

An advisory group will offer advice on implementing study findings into national guidance.

Ethics & Dissemination

The ESORT-V study was approved by the London School of Hygiene and Tropical Medicine ethics committee 14 March 2023 (Ref 27065). The study did not require UK National Ethics Committee approval as all data were pseudo-anonymised. Access to patient data in NHS Digital datasets including the NVR, HES, and ONS, without the need for patient consent has been granted by the Confidentiality Advisory Group (CAG).

Dissemination

The results of this study will be written up and published as an official report for the NIHR RfPB. It will also be published in a peer-reviewed journal and presented at international conferences.

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Table 1. Target Trial Emulation (TTE) specification for the Emergency Surgery Or noT for common Vascular conditions in the periods before and during COVID-19 (ESORT-V) study

TTE	Protocol component	Target trial specification	Target trial emulation	Sensitivity analysis
TTE-1 (Abdominal aortic aneurysm (AAA) repair)	Eligibility criteria	<i>Include:</i> - Asymptomatic AAA <i>Exclude:</i> - Endovascular aneurysm sealing (EVAS) - Complex EVAR branch / fenestrated EVAR) - Open AAA repair with anything other than infra-renal clamp site - Ruptured aneurysm - Surgery ≤ 14 days or > 150 days from threshold date - Previous open repair/EVAR - AAA max diameter > 7cm - Neck length < 10mm for EVAR - Cancer within ± 3 months threshold date	<i>Include:</i> - Asymptomatic AAA <i>Exclude:</i> - Endovascular aneurysm sealing (EVAS) - Complex EVAR branch / fenestrated EVAR) - Open AAA repair with anything other than infra-renal clamp site - Ruptured aneurysm - Surgery ≤ 14 days or > 150 days from threshold date - Previous open repair/EVAR - AAA max diameter > 7cm - Neck length < 10mm for EVAR - Cancer within ± 3 months threshold date We will exclude people that have missing surgery date, AAA max diameter, and neck length.	
	Treatment strategies	Intervention Primary open repair or EVAR from day 15-56 following threshold date Comparator Primary open repair or EVAR from day 57-150 following threshold date	Intervention Primary open repair or EVAR from day 15-56 following threshold date recorded in the NVR Comparator Primary open repair or EVAR from day 57-150 following threshold date recorded in the NVR	Change intervention treatment window to 15-84 days and comparator treatment window to 85-150 days.

	Assignment procedures	Individuals are randomly assigned to a strategy at the threshold date (the date AAA reached threshold for intervention)	Randomisation is not possible using observational NVR data. We will assign patients who receive surgery to repair the AAA between 15-56 days following the threshold date as 'early'. We will assign patients who receive surgery to repair the AAA between 57-150 days following the threshold date as 'later'. People who do not receive any procedures following the threshold date are not included in the NVR.	
	Follow-up	Follow-up start (time 0) is synchronous with study entry and randomisation (the threshold date). Follow-up ends at the earliest of 180 days, death, or end of study period.	Follow-up start (time 0) is the threshold date recorded in the NVR for each patient. Follow-up ends at the earliest of 180 days, death, or end of study period.	
	Outcomes	<i>Primary outcome:</i> - Days alive and out of hospital (DAOH) at 180 days <i>Secondary outcomes:</i> - DAOH at 1-year - Mortality at 1-year - Re-admissions - Re-interventions	<i>Primary outcome:</i> - Days alive and out of hospital (DAOH) at 180 days <i>Secondary outcomes:</i> - DAOH at 1-year - Mortality at 1-year - Re-admissions - Re-interventions	
	Causal contrasts	Intention to treat (ITT) - Analysing as randomised Per-protocol (PP) effect - Analysing as procedure actually received	Per-protocol effect - We lack data describing the decision on 'early' vs 'delayed' surgery allocation at the threshold date in the NVR data. Thus, we cannot estimate the	

		For both, we are interested in the causal contrasts in the overall study population (the average treatment effect) and in clinically important subgroups (conditional average treatment effects) Subgroups include: - Aneurysm diameter (<6 cm vs ≥6 cm) - Frailty - Sex - Age (<85 and ≥85 years) - Comorbidities	observational analogue of the ITT. We will analyse according to the procedure timing actually observed. Similar to the target trial, we are interested in the causal contrasts in the overall study population (the average treatment effect) and in clinically important subgroups (conditional average treatment effects). Subgroups will be defined using the NVR-HES-ONS linked data, and will include: - Aneurysm diameter (<6 cm vs ≥6 cm) - Frailty - Sex - Age (<85 and ≥85 years) - Comorbidities	
	Identifying assumptions	Exchangeability - We assume that the randomisation is sufficient so that the treatment and comparator groups are on average comparable on both observed and unobserved characteristics Positivity - Each participant has a non-zero probability of being assigned to the treatment or comparator No differential loss-to-follow-up	Exchangeability - We assume that the instrument in the instrumental variable analysis (see Data analysis plan) is not associated with unmeasured confounders, meaning our analysis yields treatment effect estimates unbiased by confounding Other instrumental variable assumptions: - Relevance (the instrument strongly predicts the type of surgery actually received)	

		- People remain in the trial and randomisation is preserved	- Exclusion restriction (the instrument acts on the outcome only via the type of surgery received) - Monotonicity (the instrument has a consistent impact on the type of surgery received) Positivity - Each participant has a non-zero probability of receiving the treatment or comparator (achieved through strict eligibility criteria developed by a clinical panel and confirmed via baseline characteristics) - The instrument varies across strata of the covariates No differential loss-to-follow-up - NVR-HES-ONS data capture is sufficient in that there is no differential missingness in outcome data over follow-up	
	Data analysis plan	Descriptive Baseline characteristics are compared across treatment groups to ensure randomisation balanced observed patient characteristics. Clinical effectiveness The absolute mean difference will be reported for DAOH at 180 days, and absolute risk difference for binary endpoint measures. We will report the	Descriptive Baseline characteristics are compared across treatment groups to understand potential confounding in the eligible study population. Variation in the 'tendency to expedite surgery' (proportion of people with 'early' surgery in the 12-months prior to the index date at the same vascular centre for each individual patient) will be compared across vascular centres in	

		clinical effectiveness overall, and according to pre-specified subgroups. For count data (e.g., DAOH), the outcome analysis will use generalised linear models (e.g., Poisson models). Probit or logit models will be used for binary outcomes (e.g., emergency admission). A discrete-time hazard formulation will be used for survival. Multivariable regression will be used to improve precision of estimates. Cost-effectiveness Total costs, life-years, quality-adjusted life years (QALYs), and net monetary benefit (NMB) at 1 year from randomisation will be reported. A cost-effectiveness analysis (CEA) will be undertaken to assess the relative cost-effectiveness of early surgery versus later surgery for overall study population and pre-specified subgroups. The CEA will use multivariable regressions to estimate incremental costs, incremental QALYs, and incremental net monetary benefits (INBs).	England to explore exogenous variation in the instrument. We will partially evaluate the instrumental variable assumptions by testing how strongly the instrument predicts treatment and the balance of measured confounders across levels of the instrument. Clinical effectiveness An instrumental variable analysis will be used to reduce the risk of confounding bias. A 2-stage residual inclusion (2SRI) model will be used to estimate the causal estimands of interest. The first stage will use a probit or logit model to estimate the propensity for type of surgery (early vs later) conditional on the instrument and other covariates. The second stage will use the generalised residuals from the first stage model as additional covariates in the multivariable outcome model. For count data (e.g., DAOH), the outcome analysis will use generalised linear models (e.g., Tobit models). Probit or logit models will be used for binary outcomes (e.g., emergency admission). Survival models will be used for survival data. Cost-effectiveness	
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			Total costs per patient will be reported up to 1-year using unit costs combined with resource use measures. Resource use considered as main cost drivers will be collated from NVR-HES linked dataset. Unit costs will be obtained from the NHS payment by results database⁴⁸ and the personal health services perspective research unit (PSSRU) unit cost database.⁴⁹ Life years will be calculated directly from death data. Quality adjusted life years (QALYs) will be calculated by combining each person's survival time with appropriate health-related quality of life (HrQoL) estimates from the literature and guidelines,⁵⁰ using EQ-5D as the generic instrument and the area-under-the curve approach.⁵¹ NMB will be calculated by combining cost and QALYs data. We will use the same instrumental variable analysis as in the clinical effectiveness analysis to estimate the treatment effect for the cost-effectiveness analysis outcomes (incremental costs, incremental QALYs, and INBs).	
TTE-2 (Lower limb revascularisation)	Eligibility criteria	<i>Include:</i> (A) Elective bypass with chronic limb-threatening ischaemia between 2015 to 2023.	<i>Include:</i> (A) Elective bypass with chronic limb-threatening ischaemia between 2015 to 2023.	

		(B) Non-elective (i.e., emergency) bypass with chronic limb-threatening ischaemia between 2015 to 2023 <i>Exclude:</i> <u>Patient</u> - Fontaine score less than 3 - ASA grade 5 - Aneurysm - Trauma - Uncontrolled infection - Neuropathy	(B) Non-elective (i.e., emergency) bypass with chronic limb-threatening ischaemia between 2015 to 2023 <i>Exclude:</i> <u>Patient</u> - Fontaine score less than 3 or missing in NVR data - ASA grade 5 or missing in NVR data - Aneurysm - Trauma - Uncontrolled infection - Neuropathy <u>Other</u> - Centres with low case ascertainment	
	Treatment strategies	(A) Elective Bypass: Intervention Bypass on days 1-14 from qualifying outpatient visit Comparator Bypass on days 15-60 from qualifying outpatient visit (B) Non-elective Bypass: Intervention Bypass on days 1-5 from admission Comparator Bypass on days 6-21 following admission	(A) Elective Bypass: Intervention Bypass on days 1-14 from qualifying outpatient visit as recorded in the HES data Comparator Bypass on days 15-60 from qualifying outpatient visit as recorded in the HES data (B) Non-elective Bypass: Intervention Bypass on days 1-5 from admission as recorded in NVR data	

			Comparator Bypass on days 15-60 from admission as recorded in NVR data	
	Assignment procedures	(A) Elective Bypass: Individuals are randomly assigned to a strategy at the qualifying outpatient visit . (B) Non-elective Bypass: Individuals are randomly assigned to a strategy at the time of admission (the date at which a patient has a non-elective admission requiring lower limb bypass surgery)	Randomisation is not possible using observational data. (A) Elective Bypass: We will assign patients who receive lower limb bypass 1-14 days following a qualifying outpatient visit date as 'early'. We will assign patients who receive lower limb bypass between 15-60 days following a qualifying outpatient visit date as 'delayed'. (B) Non-elective Bypass: We will assign patients who receive lower limb bypass 1-5 days following a non-elective admission for lower-limb bypass surgery as 'early'. We will assign patients who receive lower limb bypass 6-21 days following a non-elective admission for lower-limb bypass surgery as 'early'. People who do not receive any procedures are not included in the NVR.	For the Elective TTE , we will consider a narrower definition of later surgery as lower limb bypass on days 15–28 from the qualifying outpatient visit.
	Follow-up	Follow-up start (time 0) is synchronous with study entry and randomisation (the qualifying outpatient visit and admission date, for elective and non-elective TTEs, respectively).	Follow-up start is the date of the lower-limb bypass surgical procedure.	

		Follow-up ends at the earliest of 90 days (primary outcome), death, or end of study period.	Follow-up ends at the earliest of 90 days (primary outcome), death, or end of study period.	
	Outcomes	<i>Primary outcome:</i> - Days alive and out of hospital (DAOH) at 90 days <i>Secondary outcomes:</i> - DAOH at 180-days, and 1-year - Mortality at 180-days, and 1-year - Re-admissions - Re-vascularisation (index-limb) - Major amputation (index-limb) - Amputation-free survival	<i>Primary outcome:</i> - Days alive and out of hospital (DAOH) at 90 days <i>Secondary outcomes:</i> - DAOH at 180-days, and 1-year - Mortality at 180-days, and 1-year - Re-admissions - Re-vascularisation (index-limb) - Major amputation (index-limb) - Amputation-free survival	
	Causal contrasts	Intention to treat (ITT) - Analysing as randomised. Per-protocol (PP) effect - Analysing as procedure actually received. For both, we are interested in the causal contrasts in the overall study population (the average treatment effect) and in clinically important subgroups (conditional average treatment effects) Subgroups include: - Diabetes status - Fontaine score - - Kidney replacement therapy - Sex	Per-protocol (PP) effect - We lack data describing the decision on 'early' vs 'delayed' surgery allocation at the qualifying outpatient visit and admission date, for elective and non-elective populations, respectively. Thus, we cannot estimate the observational analogue of the ITT. We will analyse according to the procedure timing actually observed. Similar to the target trial, we are interested in the causal contrasts in the overall study population (the average treatment effect) and in clinically important subgroups (conditional average treatment effects).	

			Subgroups will be defined using the NVR-HES-ONS linked data, and will include: - Diabetes status - Fontaine score - - Kidney replacement therapy - Sex	
	Identifying assumptions	Exchangeability - We assume that the randomisation is sufficient so that the treatment and comparator groups are on average comparable on both observed and unobserved characteristics. Positivity - Each participant has a non-zero probability of being assigned to the treatment or comparator. No differential loss-to-follow-up - People remain in the trial and losses to follow-up are not differential with respect to the randomisation.	Exchangeability - We assume that the instrument in the instrumental variable analysis (see Data analysis plan) is not associated with unmeasured confounders, meaning our analysis yields treatment effect estimates unbiased by confounding. Other instrumental variable assumptions: - Relevance (the instrument strongly predicts the surgery actually received). - Exclusion restriction (the instrument acts on the outcome only via the surgery received). - Monotonicity (the instrument has a consistent impact on the surgery received). Positivity - Each participant has a non-zero probability of receiving the treatment ('early' surgery) or comparator ('later' surgery) (achieved through strict eligibility criteria developed by a clinical	

			panel and confirmed via baseline characteristics). - The instrument varies across strata of the covariates. No differential loss-to-follow-up - NVR-HES-ONS data capture is sufficient in that there is no differential missingness in outcome data over follow-up.	
	Data analysis	Descriptive Baseline characteristics are compared across treatment groups to ensure randomisation balanced observed patient characteristics. Clinical effectiveness The absolute mean difference will be reported for DAOH at 90 days, and absolute risk difference for binary endpoint measures. We will report the clinical effectiveness overall, and according to pre-specified subgroups. For count data (e.g., DAOH), the outcome analysis will use generalised linear models (e.g., Poisson models). Probit or logit models will be used for binary outcomes (e.g., emergency admission). A discrete-time hazard formulation will be used for survival. Multivariable regression will be used to improve precision of estimates. Cost-effectiveness	Descriptive Baseline characteristics are compared across treatment groups to understand potential confounding in the eligible study population. Variation in the ‘tendency to expedite surgery’ (proportion of people with ‘early’ surgery in the 12-months prior to the index date at the same vascular centre for each individual patient) will be compared across vascular centres in England to explore exogenous variation in the instrument. We will partially evaluate the instrumental variable assumptions by testing how strongly the instrument predicts treatment and the balance of measured confounders across levels of the instrument. Clinical effectiveness An instrumental variable analysis will be used to reduce the risk of confounding bias. A 2-stage residual inclusion (2SRI)	

		Total costs, life-years, quality-adjusted life years (QALYs), and net monetary benefit (NMB) at 1 year from randomisation will be reported. A cost-effectiveness analysis (CEA) will be undertaken to assess the relative cost-effectiveness of early surgery versus later surgery for overall study population and pre-specified subgroups. The CEA will use multivariable regressions to estimate incremental costs, incremental QALYs, and incremental net monetary benefits (INBs).	model will be used to estimate the causal estimands of interest. The first stage will use a probit or logit model to estimate the propensity for type of surgery (early vs later) conditional on the instrument and other covariates. The second stage will use the generalised residuals from the first stage model as additional covariates in the multivariable outcome model. For count data (e.g., DAOH), the outcome analysis will use generalised linear models (e.g., Tobit models). Probit or logit models will be used for binary outcomes (e.g., emergency admission). Survival models will be used for survival data. Cost-effectiveness Total costs per patient will be reported up to 1-year using unit costs combined with resource use measures. Resource use considered as main cost drivers will be collated from NVR-HES linked dataset. Unit costs will be obtained from the NHS payment by results database⁴⁸ and the personal health services perspective research unit (PSSRU) unit cost database.⁴⁹ Life years will be calculated directly from death data. Quality adjusted life years (QALYs) will be calculated by combining each person's survival time with	
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			appropriate health-related quality of life (HrQoL) estimates from the literature and guidelines,⁵⁰ using EQ-5D as the generic instrument and the area-under-the curve approach.⁵¹ NMB will be calculated by combining cost and QALYs data. We will use the same IV analysis as in the clinical effectiveness analysis to estimate the treatment effect for the CEA outcomes (incremental costs, incremental QALYs, and INBs).	
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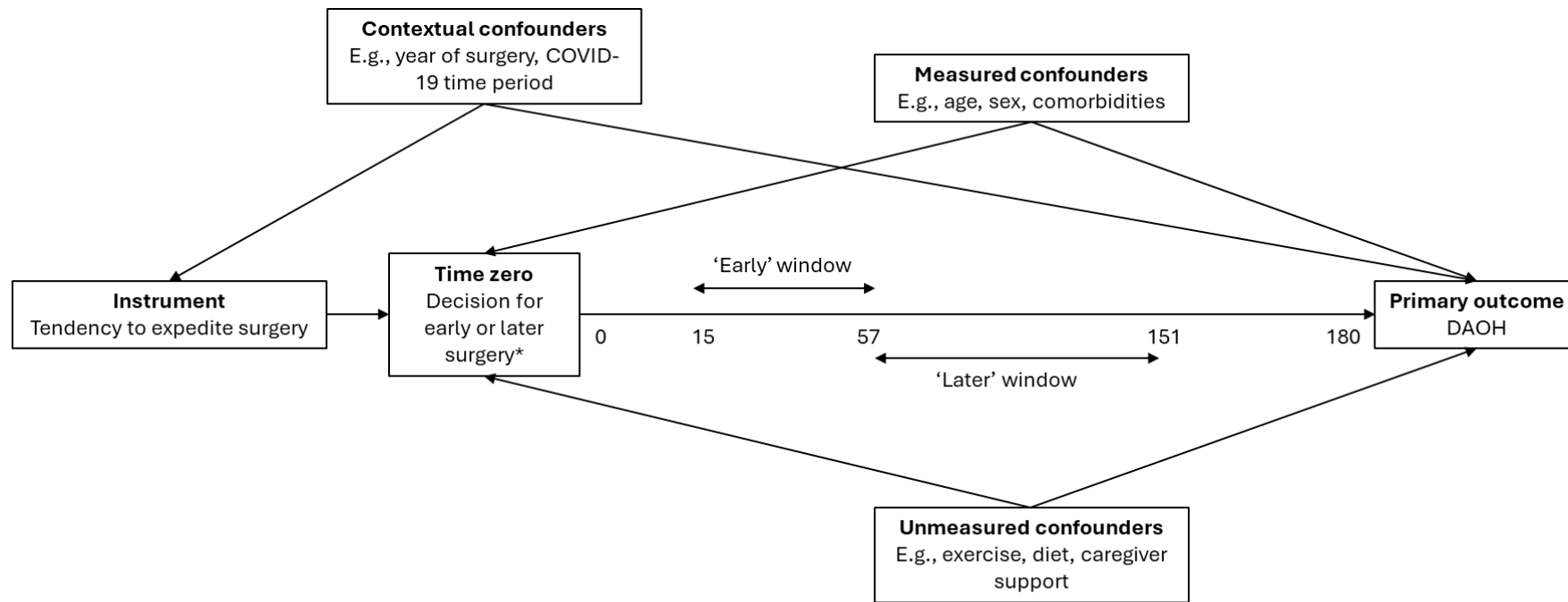
Table 2. The 4 key instrumental variable assumptions

Assumption	Details	Application in ESORT-V
Relevance	The instrument must strongly predict the procedure actually received. <i>Testable by calculating the F-statistic summarising the strength of association between the instrument and the observed procedure.</i>	People who are managed at hospitals with a higher ‘tendency to expedite’ surgery have a higher propensity to receive ‘early’ versus ‘later’ surgery.
Exchangeability	The instrument is not associated with unmeasured confounders. <i>Partially testable. We can only observe any potential associations between the instrument and measured confounders. A lack of associations will give greater confidence that the instrument is not associated with unmeasured confounders.</i>	The ‘tendency to operate’ is independent of unmeasured confounders which bias the observed associations between ‘early’ versus ‘later’ surgery and the study outcomes.
Exclusion restriction	The instrument affects the outcome only through the treatment (not via any other pathway). <i>Not testable. We will use directed acyclic graphs informed by previous research and clinical judgement to justify this assumption.</i>	Our directed acyclic graphs do not have an independent path leading from the instrument to the outcomes of interest.
Monotonicity	The instrument induces changes in the procedure in one direction only for all individuals. <i>Not testable. We cannot observe counterfactual treatment decisions for the same individual across different hospitals with different levels of the instrument.</i>	We assume that there are no defiers in the study population (i.e., that there are no patients that would get ‘early’ surgery when not encouraged by the instrument, but would not get ‘early’ surgery when encouraged).

Table 3. Measured confounders in the ESORT-V TTEs

TTE	Confounder	Definition details
1 & 2	Sex	Male or Female. Defined using the NVR.
1 & 2	Age	Defined at the admission date for surgery using the NVR.
1 & 2	BMI	Calculated (kg/m ²) using the NVR weight and height
1	AAA size	Defined in the NVR
1 & 2	Smoking status	• 1=Current or stopped within 2 months • 2=Ex-smoker • 3=Never smoked Captured by the NVR
1 & 2	ASA Grade	• 1 – Normal • 2 – Mild disease • 3 – Severe, not life-threatening • 4 – Severe, life-threatening • 5 – Moribund patient Captured by the NVR
1 & 2	Comorbidities	• Diabetes • Hypertension • Chronic lung disease • Ischaemic heart disease • Chronic heart failure • Chronic renal disease • Stroke • Lower limb arterial disease

		Captured by the NVR and supplemented with HES data
1 & 2	Frailty	• Low risk • Medium risk • High risk Defined using the SCARF Index using the HES admission mapped to the NVR index admission
1 & 2	Charlson comorbidity index	Defined using the HES admission mapped to the NVR index admission
1 & 2	Deprivation	Defined using quintiles of the Index of Multiple Deprivation (IMD)
1 & 2	Year of surgery	2015-2023
1 & 2	COVID time period	Broad definition of the COVID-19 pandemic: - pre-COVID-19 (study start to 2020 March 15) - COVID-19 (2020 March 16 to 2022 March 11) - post-COVID-19 (2022 March 12 to study end) Finer categorisation to consider based on study population size: - pre-COVID-19 (study start to 2020 March 15) - Wave 1 (2020 March 16 to 2020 May 12) - Post-Wave 1 (2020 May 13 to 2020 November 3) - Wave 2 (2020 November 4 to 2021 March 11) - Post-Wave 2 (2021 March 12 to 2021 November 3) - Wave 3 (2021 November 4 to 2022 March 11) - Post-Wave 3 (2022 March 12 to study end)

Figure 1. Simplified causal diagram for target trial emulation (TTE)-1 (abdominal aortic aneurysm repair) included in the ESORT-V study

*TTE-1: Threshold date

DAOH: Days Alive and Out of Hospital; TTE: Target Trial Emulation